# WIP: Auto Lens Upgrade

---

## PRD - Auto Lens Selection (AI-Driven)

***Mock:*** AI Tutor - Auto Lens Selection Prototype

Memo: need to include all subjects for universities, have a mimi back office for tenant type

---

### Problem Statement

Manual subject (lens) selection before snapping creates friction, 'I forgot to set the subject' moments, and slows down the "snap → help" experience.

The system should:

- Automatically assign exactly **one subject per snap** from a predefined list.
- Make the assigned subject visible and editable.
- Preserve the existing **"Explain this"** quick action under Anything.
- Allow subject changes to rerun downstream analysis without re-running OCR.
- Log manual subject overrides for analytics and evaluation.

---

### Client Context

*(Intentionally left blank.)*

---

### Competitive Analysis (Japan-focused)

#### Knock（ノック）

Camera-first homework help across subjects with follow-up interaction. Reinforces expectation that users should not configure subjects before snapping.

#### atama +

AI-driven adaptive learning system (adaptive learning = dynamically adjusting what students study next). Strong in structured subject modeling and personalization, less camera-first.

#### QANDA / Snapask

Photo-based homework help across subjects. Expectation: "take photo → get help," not "select subject → take photo."

**Photomath (International reference)**

Eliminates subject selection entirely; sets frictionless baseline for camera-first academic help.

**Implication:** Removing manual subject selection aligns Manabie with modern camera-first UX expectations while maintaining tutor-quality downstream flows.

---

## Context Flow

### Predefined Subject List (Tenant-Configurable)

- Backend defines a **closed subject list** per tenant.
- Example subjects may include:
  - English
  - Maths
  - Science
  - Social Science
  - Anything
  - Others
- The OCR agent receives this list as structured input.
- The model must classify into **exactly one subject from this list**.
- No subject outside the predefined list may be returned.

---

### Updated System Flow

Camera
→ OCR Agent (text extraction + subject classification from predefined list)
→ Lens Assignment (exactly one subject per snap)
→ Downstream Subject Agent
→ Output Generated
→ Snap Detail displays assigned subject
→ If subject is changed → rerun downstream agent chain (post-OCR only)

---

### Subject Change Rerun Logic

- OCR is **not rerun**.
- Extracted text is reused.
- Entire downstream agent chain reruns using:

  - Existing OCR text
  - Newly selected subject
  - Existing intent (solve vs explain)
- Regenerated output replaces the previous output for that snap.
- Manual subject change is logged as a trackable event.

---

## User Stories (Story + Acceptance Criteria Only)

---

#### Story 1 — Auto Subject Selection on Snap

**As a student, I want the app to automatically choose the correct subject when I snap, so I can get help instantly without thinking about categories.**

### Acceptance Criteria

- Student can snap without manually selecting a subject.
- System assigns exactly **one subject per snap** from the tenant-provided predefined list.
- “Others” is treated as a valid subject if included in the predefined list.
- If classification fails, system defaults to Anything.
- Assigned subject is visible on the capture review screen.
- Student can override the assigned subject before or after generation.

---

#### Story 2 — Preserve “Explain this” Intent

**As a student, I want to tap “Explain this” when I want understanding rather than just an answer.**

### Acceptance Criteria

- If final subject = Anything, “Explain this” quick action is visible on the review screen.
- Tapping “Explain this” routes to explanation-style output.
- Works whether Anything was auto-selected or manually selected.

---

#### Story 3 — Show Assigned Subject on Snap Detail

**As a student, I want to see what subject my snap was classified as, so I can trust or correct it.**

### Acceptance Criteria

- Snap Detail displays the assigned subject clearly.

- Exactly one subject is stored and shown per snap.
- Snap Detail indicates whether the assignment is Auto or Manual.
- Manual subject changes are logged as a trackable analytics event.

---

#### Story 4 — Change Subject and Regenerate Output

**As a student, if the subject is wrong, I want to change it and regenerate the tutor output, so the help matches my question.**

### Acceptance Criteria

- Student can change subject from Snap Detail using the tenant-provided predefined list.
- Exactly one subject can be selected per snap.
- On subject change, the system reruns the downstream agent chain using existing OCR text (OCR is not rerun).
- Regenerated output fully replaces previous output for that snap.
- Manual override events are logged and measurable.

---

## Non-Functional Requirements

- **Latency:** Auto classification must not meaningfully delay snap-to-review. Regeneration must feel responsive.
- **Determinism:** Subject must always be selected from backend-provided predefined list.
- **Reliability:** If classification fails, default to Anything. No snap or output loss.
- **Observability:**
  - Track assigned subject.
  - Track auto vs manual assignment.
  - Track manual override events.
  - Track regeneration count per snap.

---

## Out of Scope / Future Considerations

### Out of Scope

- Multiple subjects per snap.
- Per-question subject routing within a single image.
- Personalization based on student history.

- Major redesign of solver or explanation UX.

### Future Considerations

- Per-question splitting and subject routing.
- Threshold tuning driven by override rate.
- Optional exposure of subject confidence for debugging purposes.
- Adaptive subject weighting per tenant.